

A Scoping Review of Workflow Engines (Published in PubMed Before 2026) in Biomedical Research

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Abstract

The complexity of data intensive research and increasing volume have created growing need for reliable computational systems that can automate. Manual workflows are difficult to reproduce, time consuming and unreliable results with often human error. Workflow engines address these challenges by structured analysis with less human error; however, studies on workflow engines remains limited. This scoping review workflow engines listed in PubMed until 31 December 2025. Of the 126 articles, 45 articles were included – comprising 15 articles describing workflow engines, and 30 articles using workflow engines. The 15 workflow engines were analysed based on (a) source type, (b) license type, (c) programming language, and (d) operating system and computing environment. The workflow engines are varied in openness, licensing, languages, and infrastructure of the environment. These findings provide a practical reference for researchers selecting workflow systems for biomedical and research applications.

Keywords

Workflow engine, Workflow pipeline, Process engine, Process pipeline, Task engine, Task pipeline, Automation pipeline, Pipeline engine, Reproducibility, Scalability, Biomedical research.

Introduction

The growth of high throughput technologies and data intensive research particularly in biomedical research, the data volume is large and computationally complex; such as DNA sequencing and medical imaging; which has increased the demand for efficient, reproducible, and scalable computational workflows [1]. Workflow engines, such as, SGXPro [2], BioWMS [3] and Taverna [4]; have demonstrated the application of workflow system in areas such as genomic data analysis, biological database management and distributed scientific computing. These systems help organise tasks, manage dependencies, delegate resources resulting in greater efficiency, reproducibility and reducing human errors. With workflows growing more sophisticated, choosing an appropriate workflow engine require careful thought regarding each of the operational and technical characteristics [5,6]. Recent workflow engines; such as, IWEIS [7], SePIA [8] and RABIX [9]; have also emphasised the growing importance of workflow portability, containerisation and cloud native system to ensure reproducibility and scalability across diverse computational environments. Furthermore, modern workflow engines; such as, OpenMOLE [10] and Script of script [11]; increasingly support technologies; such as, Docker, Kubernetes, and distributed computing frameworks; to improve execution efficiency and facilitate collaboration among researchers.

Despite the increasing demand on workflow engine, their use remains inconsistent across implementations, accessibility and architectures. Some engines can be either stand alone or embedded as tools in larger analytic ecosystems. Additional differences

among workflow engines include whether they are open or close source, the license model, support for different programming languages and their compatibility with different operating systems. Thus, the above differences directly impact usability, scalability and integration into user current computational infrastructure [12]. Hence, this scoping review aims to compile the current state of knowledge regarding workflow engines, and to provide an overview of the technical features of workflow engines as means to aid researchers and practitioners in choosing the appropriate workflow engine to meet their computational requirements.

Methodology

A literature search was conducted on PubMed for studies published to 31 December 2025, using various combinations of terms; namely,

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(i) workflow, (ii) process, (iii) task, (iv) engine, (v) pipeline, and (vi) automation. This resulted in the following search string: ("workflow engine" OR "workflow pipeline" OR "process engine" OR "process pipeline" OR "task engine" OR "task pipeline" OR "automation pipeline" OR "pipeline engine"), with the search URL:

[https://pubmed.ncbi.nlm.nih.gov/?term="workflow+engine"+OR+"workflow+pipeline"+OR+"process+engine"+OR+"process+pipeline"+OR+"task+engine"+OR+"task+pipeline"+OR+"automation+pipeline"+OR+"pipeline+engine"&filter=-dates.1000/1/1-2025/12/31](https://pubmed.ncbi.nlm.nih.gov/?term='workflow+engine'+OR+'workflow+pipeline'+OR+'process+engine'+OR+'process+pipeline'+OR+'task+engine'+OR+'task+pipeline'+OR+'automation+pipeline'+OR+'pipeline+engine'&filter=-dates.1000/1/1-2025/12/31). Identified articles were screened using 5 exclusion criteria: namely; (A) unavailability of full text; (B) retracted article; (C) non-English article; (D) non-primary articles, such as reviews and meta-analyses; and (E) article that did not focus on workflow engine.

Results and Discussion

The PubMed search identified 126 articles; of which, 45 articles were included, and classified into 2 groups (Figure 1): (1) articles focusing on development of workflow engine (15 articles), and (2) articles focusing on workflow engine as tools (30 articles). A summary of the 15 workflow engines is given in Table 1. Of the 30 articles using the workflow engines, only 7 were using the 15 engines reviewed (Table 2). The other 23 articles using the workflow engines, summarized in Table 3, were using workflow engines not published PubMed.

Although the law embodies specific ethical valuations through normative codes, concrete reality frequently presents unprecedented dilemmas, meaning that ethics substantially exceeds the articulation of current legal frameworks.

Description of the 15 Workflow Engines

This section presents a description of each of the 15 workflow engines. Snakemake [12] is a workflow management system that features a Python-like language for creating workflows and features an execution environment that scales from personal computers running on one core to computing clusters while not altering the workflow.

One of its distinguishing features includes support for automatic inference of multiple named wildcards in input and output file names. At same time the platform is highly scalable. These features have made its adoption mainly in bioinformatics and data-intensive scientific research. The Snakemake workflows were inspired by the GNU Make.

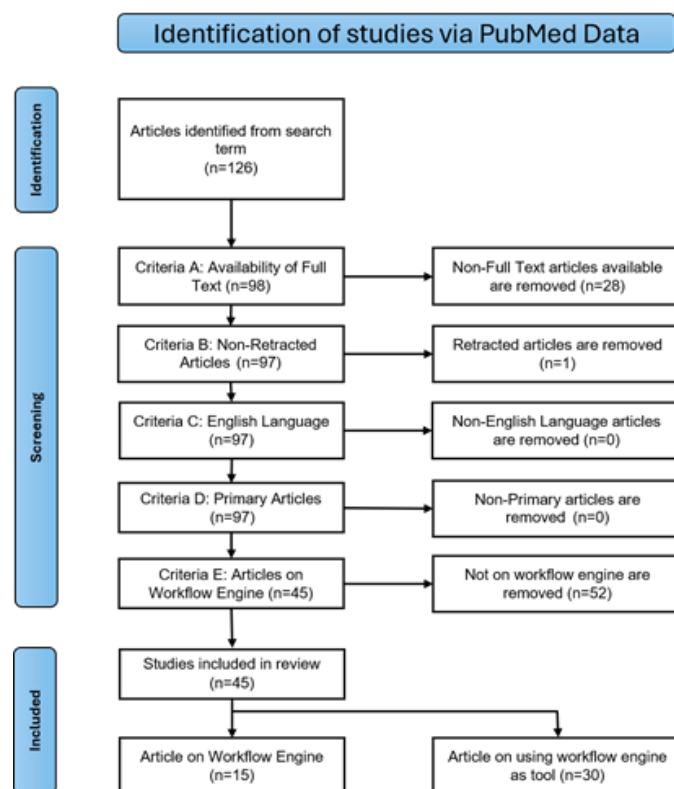


Figure 1: PRISMA Flowchart [13].

Table 1: Summary of Workflow Engines.

Engine	Repository	Licence	Language	Operating System
Snakemake [12]	https://snakemake.github.io/	3-clause BSD license	Python	Linux, MacOS
DEWEY [14]	Closed source	Not stated	Java, C++, python	Linux, windows
TomoBEAR [15]	https://github.com/KudryashevLab/TomoBEAR	GNU General Public License version 3	Multi language	web based server
SePIA [8]	https://github.com/SEPIA-Framework/sepia-stt-server	GNU General Public License	Java and multi-language	HPC cluster
SGXPro [2]	Open source but unable to access	Not stated	Not stated	HPC cluster
Scipion3 [16]	https://github.com/scipion-em	GNU General Public License	Python and C++	HPC cluster and GPU nodes
Script of script [11]	https://github.com/vatlab/SoS	BSD 3-clause	Multi language	Multicore
BioWMS [3]	Open source but unable to access	Open academic release, BMC journal	Java	Web server
HSWAP [17]	Closed source	Not stated	Java and python	HPC cluster
Clinical Decision-Support Workflow Engine [18]	http://sourceforge.net/projects/sharkwf	GNU General Public License v3	Java / XPDL	Web server
RABIX [9]	http://github.com/rabix/bunny	Apache 2.0 License	Python, JAVA	HPC, local, cloud
Conveyor [19]	Open source but unable to access	GNU General Public License	Java	Multicore
OpenMOLE [10]	https://openmole.org/	AGPL license	Scala/DSL	HPC cluster
IWEIS [7]	Closed source	No stated	Java	Web server
Taverna [4]	https://taverna.apache.org	LGPL license	Java	Web server

Table 2: Applications of Workflow Engines Review in this Study. Seven of the 30 articles on applications of the 15 workflow engines (listed in Table 1) are briefly described here.

Engine	Brief Application Notes
Snakemake [12]	Used to automate rule-based bioinformatics pipelines, connecting inputs, outputs and dependencies across analytical steps. In the reviewed papers, snakemake drives the GBS-DP pipeline to handle large genotyping by sequencing datasets. It also powers SnakeChunks, supporting quality control, mapping and comprehensive bacterial ChIP-seq and RNA-seq analysis [20,21].
DEWEY [14]	Applied for the automation of the radiology department. It receives images, classify them, initiate processing tasks, controls human worklists and route final output as image or result for clinical review [14].
TomoBEAR [15]	Used in the processing of cryo-electron tomography data such as tomogram reconstruction and subtomogram averaging to determine macromolecular structures [15].
SePIA [8]	Used to analysis total RNA-seq, small RNA-seq and integrated miRNA-mRNA. It is based on Andruil engine, which handle preprocessing, mapping, expression quantification, differential-expression analysis, functional analysis and HTML reporting [8].
SGXPro [2]	Help for the determination of the 3D structure of proteins by X-ray crystallography, developed as a parallel system. It includes crystallographic programs, allows communications and conversion of files between programs, and tests a variety of algorithms and parameter sets to find the most appropriate solution to the structure [2].
Scipion3 [16]	Among the reviewed papers, Scipion is applied to both structural biologists and image analysis. Applications include cryogenic-EM image analysis through ScipionTomo, virtual drug screening through Scipion-Chem, structural and dynamic analysis through Scipion-EM-ProDy [22-25].
Script of script [11]	Used to enable computational research with several programming languages such as python, R and shell script. It allows researchers to write scripts and run them on one or more computers both locally and remotely, on high-performance computers [11].
BioWMS [3]	Applied to the definition, execution and monitoring of experiments in bioinformatics. It dynamically creates an agent-based workflow engine, which can be executed in a distributed fashion, re-used and managed workflow intermediate and final results [3].
HSWAP [17]	It combines modelling, simulation and data analysis software and has been used in a high-throughput molecular screening of energetic materials [17].
Clinical Decision-Support Workflow Engine [18]	Applied in clinical decision-support logic. It enables clinical rules to be tested with retrospective EHR data and then deployed to react to real-time clinical events [18].
RABIX [9]	It facilitates the interpretation and coordination of workflows to support biomedical or genomics analysis over a computing environment, allowing for better reproducibility, portability and reusability [9].
Conveyor [19]	Used for building, running and tracking large bioinformatics pipelines. It is concurrent, can handle loops, branches and other control structures. It has been shown to be usable in genome annotation and comparative-genomics applications [19].
OpenMOLE [10]	Applied to the large-scale exploration of computational models, distributes model simulations across clusters, grids and cloud-computing environments, and distributes parameter exploration and optimisation experiments [10].
IWEIS [7]	Applied to imaging-based clinical trials. It gives the study coordinator the flexibility of creating data collection, image-management, analysis, approval and notification processes without excessive software [7].
Taverna [4]	Taverna offers an application with analytical tools, databases and web services can be combined in user friendly ways into re-usable scientific workflows and that has, in the appears accompanying the CDK-Taverna, it integrated the CDK for computing chemical descriptors, processing compounds, analysing structures and cheminformatics workflow using metabolomics or drug discovery experiments [26].

Table 3: Applications of Workflow Engines Not Review in this Study. This table briefly describes the remaining 23 articles on workflow engine usage (from the 30 articles on workflow engine usage), which are not based on the 15 workflow engines listed in Table 1.

No.	Application	Workflow Engine Used
1	Applied to reproducible biomedical analyses, including differential gene-expression analysis and somatic-mutation calling across different computational environments [27]	CWL/cwltool (RRID:SCR_015528)
2	Applied to coordinate lightweight microservices in distributed computing contexts to bridge and control computations in the cloud, at the edge, and at the device level [28]	Astermule/Astertower
3	Applied to large-scale neuroimaging studies by coordinating imaging datasets and submitting analysis jobs to HPC [29]	DAX-1 http://dax.readthedocs.io/
4	Applied to systems-biology research activities including omics-data analysis, biological modelling and simulation of complex biomedical systems [30]	BioUML https://www.biouml.org/
5	Applied to a local-ancestry aware genome wide association study combining ancestry inference, haplotype processing, and association testing [31]	Nextflow/tractor pipeline (RRID:SCR_024135)

6	Applied to bacterial whole-genome sequencing to manage quality control, genome assembly, annotation, and antimicrobial -resistance detection [32]	Nextflow/BALROG-ISO (RRID:SCR_024135)
7	Applied to scientific workflows on HPC systems to manage workflow-generated data, its provenance, and run-time information [33]	d-chiron/schalaDB https://github.com/hpcdb/d-Chiron
8	Applied to next generation sequence (NGS) workflows including sample management, demultiplexing, quality control, mapping, and downstream bioinformatics analyses [34]	SMITH/Galaxy (RRID:SCR_006281)
9	Applied to perform sequencing and variant analysis for the ST8SIA2 and NCAM1 genes as an attempt to identify genetic variants associated with bipolar disorder [35]	Galaxy (RRID:SCR_006281)
10	Applied to identify in-complete radiology exams and automatically notify clinicians to support follow up of misses in diagnostic test services [36]	soarian workflow engine
11	Applied to routinely acquired head MRI examinations to automate various examination steps thereby shorten examination time and reducing operator intervention [37]	Siemens Dot
12	Applied to semiautomated glycopeptide mass spectrometry analysis including spectrum processing, candidate mapping, and glycopeptide identification [38]	TOPPAS/OpenMS (RRID:SCR_000533)
13	Applied to automated downstream analysis of large-scale data including protein/metabolite quantification, identification, and visualisation [39]	KNIME/OpenMS (RRID:SCR_006164)
14	Applied to automate high-throughput assembly of large synthetic DNA libraries and thereby to reduce manual laboratory handling DNA construction phase [40]	Assemblatron
15	Applied to personalised prostate cancer screening automatic delivery of individualised reminder messages via SMS, email and automated voice messaging [41]	Rule-based workflow engine
16	Applied to provide real-time web-based monitoring, visualisation, and interaction with scientific workflows [42]	kepler/webview (RRID:SCR_005252)
17	Applied to implement executable clinical guidelines and screening processes for supporting clinical decision support [43]	Bonita
18	Applied to a national wide blood-transfusion practice evaluation programme, to track activities, submitted evidence and progress of participants [44]	J@LON workflow system
19	Applied to microscopy and neuroinformatics workflows, connecting image acquisition, processing, storage, and neuroscience data resources[45]	Kepler (RRID:SCR_005252)
20	Applied to large scale bioinformatics analyses through co-ordinates SOAP web services and partitioning large bioinformatics data sets [46]	BPEL workflow engine
21	Applied to manage and distribute medical images across clinical departments beyond radiology and for enterprise-wide access to imaging information [47]	PACS workflow system
22	Applied to a proactive of remote monitoring of patients suffering from chronic disease and to co-ordination of health-related information between the patient, caregivers and clinicians [48]	Healthcare workflow engine
23	Applied to distributed 3D liver perfusion mapping from MR-DCE images to enable computational image-processing task parallelisation [49]	MOTEUR/Gwendia

DICOM-Enabled Workflow Engine System (DEWEY) [14] is a workflow engine, defines how the workflow engine can be deployed to improve efficiency in the radiology department of Mayo Clinic. The system is designed considering scalability, reliability and flexibility, which involves the use of many workflows, with one workflow replacing the existing manual workflow. The architecture is made up of five core modules, namely: DICOM receiver/sender, message queue, database, workflow engine (Activiti), and compute server. DEWEY constantly evaluates the output from Magnetic Resonance Imaging (MRI) and Computed Tomography (CT) machines, routing the scans into the change detection and CAD algorithms, thus improving the efficiency of the examination conducted in the set period compared to human work.

Electron Tomography and Automatic Reconstruction (TomoBEAR) [15] workflow engine is modular and configurable, enabling users to optimize their workflows for subtomogram averaging with cryo-electron tomography datasets. It includes a set of commonly used cryo-Em software with sensible defaults, creating a transparent "white box" approach to handling and analysing data. This system contains more than 20 models, combining popular cryo-ET software programs like IMOD, Dynamo, MotionCor2, AreTomo, and GCTF/CTFFIND4, as well as modern machine learning algorithms such as IsoNet and crYOLO. This system has been shown to be effective on structural

biology studies involving protein complexes and cellular specimens such as RyR1 ion channel in its native lipid bilayer environment, and Plasma Focused Ion Beam - milled lamellae, yielding structures at high resolutions with less manual input.

Sequence Processing, Integration and Analysis (SePIA) [8] is a workflow engine for processing and analysis of small RNA and RNA sequences, which allows the execution of more than 20 proven RNA-seq software directly upon an approved workflow engine. It is a flexible pipeline design is capable of supporting the handling, analysing and integration of both small RNA and RNA sequence data. The implementation of SePIA, using Docker framework increases its portability and usability, and the outcome data is presented in an easily navigable HTML report. SePIA as a highly modular framework that facilitates customisable experiments in order to maintain the same workflow framework.

Southeast Collaboratory for Structural Genomics Protein (SGXPro) [2] is a parallel workflow engine which is designed specifically for automated structural determination using X ray crystallography. The program incorporates several software packages in one single environment and allows the combination of algorithms without the need to worry about compatibility issues and file formats. Automated parameter search space optimization can be performed using this tool

along with its client server approach to take advantage of distributed computing. Plug and play functionality of the software allows easy incorporation of newer programs for future research needs.

Scipion3 [16] is an engine which serves as a workflow platform for cryo-EM data analysis and structural biology research. This software provides a common platform for combining various imaging algorithms within reproducible pipelines. Modularity is considered the key aspect of Scipion3, which allows scientists to use different algorithms in one analysis and maintain its traceability and reproducibility at the same time. In addition, Scipion3 improves the user interface and automation options, which helps with the efficient processing of large volumes of data associated with cryo-EM experiments.

Script of Scripts (SoS) [11] is a workflow system aimed at simplifying computational research by incorporating different scripting languages into one single tool. It makes it possible to combine interactive data exploration and analysis with structured workflow processes, making it easier to plan and run complicated computations efficiently. Some of its key features include multi-language scripting, dynamic reporting and execution on various platforms, including high-performance computer clusters. With the versatile syntax options that it offers, workflows can be defined in a process-oriented manner or in an outcome-oriented manner. This SoS workflow system is highly useful for exploratory research because of its ease to use, readability and reproducibility.

BioWMS [3] is a web-based workflow management system for supporting in-silico bioinformatics experiments. BioWMS is designed to allow to define, execute and monitor workflows with standardised process definition languages. BioWMS provides an interface for scientist to define and enact workflows through its web interface. BioWMS is designed to be compositional, allowing workflows to use other workflows as components. BioWMS was developed at the University of Camerino to reduce the technical barrier for biologist to use complex data analysis pipelines.

HPC Scientific Workflow Application Platform (HSWAP) [17] is a system which utilises an automated stimulation platform that leverages workflow engine technology to facilitate molecular screening simulations in scientific research. The platform combines different computational programs to execute complicated simulations effectively. The use of workflow engine technology enables coordination of task scheduling and execution. The design allows easy addition of new workflows and programs in the future. It can perform simulations at high-performance levels. However, the article lacks detailed information on implementation details, such as coding languages, software licenses, or open source, so there is limit reproducibility.

The application of Clinical Decision-Support Workflow Engine [18] technology to the domain of clinical decision support (CDS), specifically with electronic health records (EHR). The authors implemented an open-source workflow suite using XPDL, which allows users to graphically model clinical decision logic. Major contribution of the author is the retrospectively testing of the decision logic against patient data, which is a major challenge in the implementation of CDS technology. The system designed as user-friendly, flowchart-based representation of the knowledge, which is vendor-independent, allowing for the potential integration of the system with any EHR system.

RABIX [9] is a workflow execution engine for the common workflow

language, which is a new standard for the description of portable and reproducible tools and bioinformatics workflows. The research emphasises the problem of reproducibility in the analysis of large biomedical data sets and proposes workflow executors as a potential solution to the problem. RABIX with CWL, features error logging, file management, optimization of job scheduling, and graph transformation of nested workflows, which improves the workflow execution process. By allowing downstream tools to begin execution as required inputs are available, RABIX improves the overall workflow runtime.

Conveyor [19] is described as a multi-layered generic bioinformatics workflow engine intended for workflow composition, execution as well as monitoring. This tool is said to exhibit an open architecture that is extensible in nature. Such an architecture is able to allow the execution of tasks in a concurrent fashion. This is particularly significant in order to effectively utilise modern CPU multi core technology. Unlike other simple workflow pipelines, it able to allow branches as well as loops in workflows. such characteristics enable it to support complex workflows. The workflow is said to be decoupled from workflow execution management. This is particularly significant in order to enable researchers to effectively focus on biological analyses without concerns related to infrastructure. Two examples of bioinformatics workflow were presented in this paper to illustrate the flexibility of the workflow engine.

Open Model Experiment (OpenMOLE) [10] is a workflow management system for the conduct of reproducible large-scale neuroimaging research. OpenMOLE workflow management system has the capacity to support the development of a domain specific language, which allows for the seamless distribution of computations across various computing environments with ability such as desktop environments and large high-performance computing environments, without need for modifications. The application of the workflow management system has been demonstrated for the conduct of brain image analysis research, showing its capacity for the conduct of large-scale neuroimaging research involving thousands of individuals

Imaging Workflow Engine Informatics System (IWEIS) [7] is a web-based imaging informatics system that has been developed to support the management of imaging based clinical trials. It helps project coordinators to develop infrastructures for managing data in accordance with the study protocol without programming knowledge using its visual workflow designer. It also employs WADO for managing images and was implemented in the context of a rehabilitation clinical trial. It can lead to significant reductions in costs compared to developing a system from scratch. This tool was published in "Computers in Biology and Medicine" in 2015 and is intended for researchers involved in clinical trials using images who do not have access to software engineering expertise

Taverna [4] interaction service is an extension to the scientific workflow system Taverna, intended to formalise and enable the integration of human interaction within otherwise largely automated workflows, acting as a bridge between automated workflow engine and human users, it introduces a pause at predetermined points in the workflow to solicit human input before proceeding. This is a fundamental requirement for workflows that involves processes requiring expertise, or decision-making processes that cannot be fully automated. Both the Taverna software and the plugin for the interaction service are available for free and open source. This study is a key step towards filling a critical gap in the complete automation of complex life science research processes.

Availability of Repository

Analysis of the 15 articles describing workflow engines revealed that majority of the workflow engines in the articles are open sources. The identified open-source workflow engines are Snakemake [12], TomoBEAR [15], SePIA [8], SCIPION3 [16], Script of script [11], RABIX [9], OpenMOLE [10] and Taverna [4], these platforms provide accessibility of their source code. Other workflow engines, including SGXPro [2], BioWMS [3] and Conveyor [19], were described as open source in the article, but publicly accessible source code or URLs were not available within the reviewed articles. Whereas DEWEY [14], HSWAP [17] and IWEIS [7] were identified as closed source workflow engines. These sources are maintained by specific company and have exclusion right, where restricted for public access of the source code, thereby limiting opportunities for modification.

License Type

Among the 15 workflow engines, GNU General Public license (GPL) was most often used license in the workflow engines such as TomoBEAR [15], SePIA [8], Scipion3 [16], and Conveyor [19]. Snakemake [12], and SoS [11] are used permissive BSD 3-clause license, while RABIX used Apache 2.0 license. Other licensing models are LGPL, and AGPL. Several workflow engines including, DEWEY [14], SGXPro [2], HSWAP [17], and IWEIS [7] were no report license information in the reviewed articles.

Programming Language

Analysis of these articles identified that workflow engines utilised a variety of programming languages. Java was the most reported programming language and was used in the workflow engine such as SePIA [8], BIOWMS [3], Conveyor [19], IWEIS [7] and Taverna [4]. Python was the second most used programming language in the workflow engines such as Snakemake [12] and SCIPION3 [16]. Several workflow engines use multiple programming such as python and java combination. For example, DEWEY [14], HSWAP [17] and RABIX [9], to support core management and execution. Other workflow engines include TomoBEAR [15], SoS [11] and OpenMOLE [10], uses alternative languages, such as multi language, Scala, C++, and DSL (Domain-specific languages). Overall, the diversity of programming languages suggests that workflow engine development is influenced by the computational requirements and target user communities of each platform [50].

Operating System and Computing Environment

SePIA [8], RABIX [9] and OpenMOLE [10] are the workflow engines were designed within HPC (high performance computing) environments to handle and calculate massive dataset such as DNA replication. HPC clusters was the most supported operating system among the reviewed workflow engines. Other workflow engines, such as Snakemake [12] and TomoBEAR [15], were supported by LINUX operating system and web-server infrastructure. In addition, others workflow engines provided with multiple operating systems, including windows, MacOS and multicore. Thereby, increasing accessibility across diverse computational settings.

General Discussion

This review investigated workflow engines featured in scientific studies. The review revealed there is substantial heterogeneity among workflow engines. There is one thing in prevalent in all is to aim to make workflow automation smoother, boost reproducibility, and help scientists handle computational tasks efficiently [5].

Open source was the majority, 12 out of 15 engines listed open

source. Open-source software facilitates transparency, encourages collaboration, and allows researchers or developers to modify it, customise, and check analytical logics; which is crucial in biomedical fields such as bioinformatics and computational biology. Whereas close source software, restrict the code for public access and make it harder for modification and independent verification of the results. However, some engines stated open source did not have clear repositories, showing that claims of openness do not means practical access [51,52].

Most of the reviewed workflow engines used GNU General Public License (GPL) variant. With the GPL license, any users can access, modify and share the source code. Crucially, GPL have interesting nature called copyleft, that ensures any changes or improvements under open source, even if the software gets redistributed. This keeps future versions accessible and useful for researchers. There is another reason GPL license used more on academic settings. Mostly researchers focus on sharing and validating knowledge. They want tools that has ability to check, adapt and improve. GPL licensing lines up these values, helping research groups build on each other's work and allows verification of results [53]. In contrast, few workflow engines use permissive licenses like BSD, Apache, which allows broader use and developers can fold open source into closed source with minimal restrictions. While it made integration and commercial use easier, but it does not suit always for academic research. Thereby, in terms of transparency and reproducibility, GNU General Public License (GPL) variant tends to be license of choice than another license [52].

Java and Python are most often used languages in workflow engines. Java and python have carved out solid reputations in scientific software. Java was common and enterprise level reliability due to its independence, strong memory management, and durability for long- running tasks. Python is a straightforward syntax, adaptable and popularity in bioinformatics, make it more ideal for rapid development in workflow tools. Some engines have multi language, blending different strengths for strong execution and flexibility. These findings show that the choice of language depends on accessibility and specificity of the needs.

Most workflow engines favour HPC (High-performance computing) and LINUX environments to run. HPC infrastructures are built to support the handling of massive datasets such as DNA replication and complex, compute-heavy tasks. Biomedical research is generating more data such as genomics, proteomics, medical imaging. So, engines are favourably look for scalability. Other hand, Linux is everywhere in academic research due to stability but with minimal data. Cloud environments and HPCs clusters are emerging future pathways. There are some engines that also support Windows and macOS give researchers more options, especially for non-computational background users [50].

The workflow engines reviewed showed different levels of specialisation. Some include Snakemake [12], Taverna [4] and OpenMOLE [10], which cover a wide range of scientific applications and work as general-purpose engines. Others, such as TomoBEAR [15] and SCIPION3 [16], focus on much narrower fields, targeting cryo-electron tomography and cryo-electron microscopy respectively. Engines built for a specific area usually offer better features and are more finely tuned for their niche, but they do not always transfer well to unrelated research. So, choosing a workflow engine should not depend only on its technical specifications. Although need to look closely at what project needs [54].

Overall, across the 4 categories, openness lean towards open source and license, whereas choice of language and computing environment to run is depending on needs of the task. Thereby, workflow engines have become the backbone of computational research and suggest that no single engine fits every scientific project. The choice should depend on the research area, needed analysis tools, open source, reliance on Java and python, compatibility with HPC and LINUX and usability, reflects how research needs are changing and emphasis on transparency, reproducibility, and scalability. Therefore, frameworks developed in this review serves as a practical starting point for making these decisions.

Recommendations for Future Workflow Engine Development

Future research on workflow engines may focus on benchmarking performance, resource use, scalability, and user experience. Although, expanding research more on emerging technologies, such as cloud computing and Artificial intelligence and how they can be used to make workflow engines better. For example, AI powered workflow provenance analysis and dynamic runtime adjustments are active research areas. Cloud-HPC hybrid models also offer promising avenues to balance resource availability, latency and data mobility [55].

Limitations of the Review

This review has few limitations. Firstly, the search conducted only on PubMed database, which have excluded other databases such as IEEE Xplore, ACM Digital Library, and Scopus. This explains why 23 of the 30 articles that uses workflow engines that are not reviewed in this study. Second, the included article size is small, as it may vary if large articles were explored. Thirdly, some of the articles did not state the information, it affects the consistent of the study. Lastly, no software quality assessment and benchmarking performance were done. Consequently, the findings describe reported features rather than comparative performance.

Conclusion

Workflow engines have turned into most wanted tools for handling today's complex, data heavy scientific workflows. This review uncovered a difference between engines from source types, what programming languages they use, and which environments they can run in. Most workflow engines are open source, which shows researchers focus more on transparency and reproducibility. Final principle is that workflow choice must be context specific and area of interest. Thereby, my study gives a structure guidance on technical features of the workflow engines with areas support more on reproducible and scalable however, in future require a benchmarking performance for stronger decision.

Supplementary Materials

Supplementary materials can be downloaded from https://bit.ly/WorkflowEngines_SR.

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